MIL-C-45617A(MU)
9 NOVEMBER 1962

SUPERSEDING
MIL-C-45617(ORD)
23 APRIL 1959

MILITARY SPECIFICATION

COMPUTER, BALLISTICS: M13A1

1. SCOPE

1.1 This specification covers one type of electro-mechanical computer known as Ballistics Computer: M13A1", hereinafter referenced as "computer".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	—Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-C-45128	—Computers; Ballistics: M13 and M13A1; Packaging of
MIL-I-45208	—Inspection Requirements, General Specification for
MIL-I-45607	—Inspection Equipment, Supply and Maintenance for Ordnance

STANDARDS

MILITARY

MIL-STD-105	—Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	—Identification Marking of U.S. Military Property

DRAWINGS

MUNITIONS COMMAND

F8585600	—Computer, Ballistics: M13A1
IEL-8585600	—Inspection Equipment List for Computer, Ballistics: M13A1
LSQAP-8585600	—List of Supplementary Quality Assurance Provisions for Computer, Ballistics: M13A1

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Initial production sample.** Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 **Materials.** Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 **Construction.** The computer shall be manufactured in accordance with Drawing F8585600 "Computer, Ballistics: M13A1" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired electrical and mechanical

FSC 1220

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characteristics, performance and permanence to assure inherent quality and performance capabilities.

3.4 General specifications. The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL–F–13926.

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Workmanship

3.5 Environmental.

3.5.1 *Storage temperature.* The computer shall be exposed to ambient temperatures of minus 80° and plus 160° Fahrenheit until thermal equilibrium is reached. After thermal equilibrium has been reestablished at standard ambient temperature (plus 60° to 90° F.) the comuter shall show no evidence of physical failure such as casting failure or glass breakage, and shall meet the requirements of 3.6.

3.5.2 *Operational temperatures.* The computer shall be exposed to ambient temperatures of minus 40° and plus 150° Fahrenheit until thermal equilibrium is reached at each temperature, the computer shall then be inspected at the above temperatures to the following requirements:

(a) Applicable requirements of 3.14 at temperatures indicated in table II.
(b) At minus 40° Fahrenheit the outer index shall follow the inner index within one-half a graduation width over the full excursion of the range scale in a total elapsed time of 8 seconds for each direction when checked with 18 plus or minus 0.5 volts direct current (dc).
(c) At plus 150° Fahrenheit, using 28 plus or minus 0.5 volts, dc, with the ammo selector set on the AP–T, M318A1 Cam, with a load of between 14 and 18 ounce inches applied to the output shaft for the ballistics drive, and with the indexes separated 4,000 yards, the outer index shall stabilize on the inner index and shall meet the applicable requirements of 3.13 when switch is thrown to the "on" position.

3.6 Post environmental check. Each computer which has been subjected to the environmental conditions of 3.5 shall meet the following requirements after thermal equilibrium has been established at standard ambient temperature:

(a) Applicable requirements of 3.13 and 3.14.
(b) The following requirements of table IV when checked under power operation at 28 plus or minus 0.5 volts with 75 to 90 ounce inch load applied to the output shaft for ballistics drive.

AP–T, M318A1	500, 1,500 yards
HEAT, T108	500, 4,800 yards
HEP, T142E3	3,000, 3,600 yards
HVAP, M332	500 yards

The 18 volt check and hand check to requirements of table IV are not required, post environmental check of spare cams is not required.

3.7 Sealing. The interior of the computer shall show no evidence of water penetration after being subjected to a simulated rain fall as required by the rain test as specified in Specification MIL–F–13926, Procedure I. A time duration of fifteen minutes shall be substituted for two hours.

3.8 Drying. Immediately upon completion of the requirements of 3.5 and 3.7 the interior of the computer shall be dried by clean dry warm air of approximately 100° to 140° Fahrenheit.

3.9 Vibration. With the ammunition selector set on the HEP–T–142E3 Cam, the outer index set at 3,600 yards, the computer (less any shock mounts and brackets, Universal Assembly B3200862 and Output Shaft

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B8200801) shall be vibrated at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second for duration of five minutes plus or minus fifteen seconds. Subsequent to vibration the computer shall not be damaged or show evidence of broken, loose or missing parts and it shall be capable of meeting all requirements of this specification.

3.10 Manual operation.

3.10.1 *Input shaft.* Input shaft rotation shall be measured from a precise 1,500 yards check point previously determined by the manufacturer and indicated by an index line inscribed on the input shaft in coincidence with an index on the input shaft seal cover plate.

3.10.2 *Output shaft.* With the computer having a constant load of between 75 and 90 ounce inches applied to the output shaft for ballistics drive, set the ammunition selector on the AP–T, M318A1 Cam. With the range correction knob set to zero and with the input shaft set at the 1,500 yard check point, actuate the superelevation hand crank to match the outer index with the inner index within plus or minus half a line width by approaching this range first from one direction then from the opposite direction, without over travel. Note the position of the output shaft for ballistics drive and split the backlash error at the output shaft. This is then the check point for computer output. The position of the superelevation counter shall be established from this check point and shall conform to the requirements of table I.

TABLE I. *Position of Superelevation Counter*

Position of output shaft for ballistics drive	1–R 72° CCW	1,500 Yards check point 0°	2–R 72° CW	6–R 72° CW
Reading of superelevation counter in mils	3.00	9.0	20.00	40.00
Tolerance on position of output shaft for ballistics drive	±8°	±3°	±8°	±8°

3.10.3 *Outer index and inner index.* Using the AP–T, M318A1 Cam only, position the input shaft to correspond with each of the ranges as indicated in table IV and rotate the hand crank until the outer index is in coincidence with the inner index, at which point the rotation of the output shaft for ballistics drive shall correspond to the value in table IV. This procedure shall be repeated to check each of the seven other cams at 1,500 yards only. All checks shall be made by approaching the index from clockwise and counterclockwise directions.

3.10.3.1 *Backlash.* Backlash of the output shaft for the superelevator shall not exceed a total of one degree at any point of rotation of the shaft.

3.10.4 *Outer index and range scale.* Using AP–T, M318A1 Cam only, rotate the hand crank until the outer index is in coincidence with the selected range on the computer range scale for each of the ranges as indicated in table IV, at which point the rotation of the output shaft for ballistics drive shall correspond to the values in table IV. All checks shall be made approaching the index from clockwise and counterclockwise directions.

3.11 Servo follow-up sensitivity. With an output shaft load of between 75 and 90 ounce inches applied to the output shaft for the ballistics drive, ammunition selector set for the HVAP, M332 Cam, the power on at 18 plus or minus 0.5 volts and the automatic followup cycle completed at ranges of between 4,000 and 4,800 yards, rotate the input shaft slowly so that it is possible to count the number of times the contact closes in order to advance the output shaft to ballistics drive by approximately 72°. Calculate the average change in superelevation per one make and break of the contacts. Repeat in the opposite direction. The aver-

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age allowable superelevation increment for each range shall not exceed 10° 45′ and the maximum superelevation increment shall not exceed 21° 30′ as read at the output shaft to ballistics drive.

3.12 **Power operations.** With the computer having a constant load of between 75 and 90 ounce inches applied to the output shaft for the ballistics drive, the rotation of this output shaft shall correspond to the values indicated in table IV for the ammunition and ranges specified therein. The test shall be run by allowing the outer index to approach the inner index from both increasing and decreasing range with power on continuously at 28 plus or minus 0.5 volts. The computer shall also meet the accuracy requirements of table IV for the following ammunition and at range indicated when checked at 18 plus or minus 0.5 volts.

Ammunition and ranges for 18 volt check

HEP, T142E3	1,500,	3,000	and	3,600 yards
APC–T, M82	1,500,	4,000	and	4,800 yards
HEAT, T108	1,500,	4,000	and	4,800 yards

All checks shall be made by approaching index from both clockwise and counterclockwise direction.

3.13 **Stability.** Using 28 plus or minus 0.5 volts with a load of between 14 and 18 ounce inches applied to the output shaft to ballistics drive and with ballistic correction set to introduce minus fifteen percent within one percent, the rotation of the input shaft between minimum range and 1,500 yards shall not cause the computer servo to display a self-sustaining or oscillatory operation when checked on any of the six cams within the computer. This is also required for all spare cams accompanying the computer. A maximum of eight changes in direction of the servo motor, as a settling out action, shall be considered as satisfactory operation. These eight changes may be counted as clicks.

3.14 **Operability.**

3.14.1 *Torque.* The maximum operating torques shall not exceed the values given in table II at temperatures indicated.

TABLE II. *Operating Torque Values*

Control	Standard ambient temperature and plus 150°F.	Cold temperature (minus 40°F.)	Operation
Input shaft	16 oz. in.	45 oz. in.	Rotation.
Correction Knob.	34 oz. in.	45 oz. in.	Rotation.
Correction Knob.	20 lbs.	24 lbs.	Pull out force.
Ammo. Selector.	60 lb. in.	80 lb. in.	To lift follower.
Ammo. Selector.	10 lbs.	20 lbs.	To translate followers.
Hand Crank.	6 lb. in.	20 lb. in.	To rotate while engaged.
Hand Crank	12 lbs.	20 lbs.	To hold in engagement.
Interrupter Switches.	Mechanically & Electrically Operable.	Mechanically & Electrically Operable.	In accordance with 3.5.
Reset Button.	15 lbs.	40 lbs.	In accordance with 3.5.

3.14.2 *Time of response.* At standard ambient temperature with the load as specified in 3.12 applied, the inner and outer indexes positioned to the values of table III, and the ammo selector set on the HEAT, T108 Cam, the outer index shall stabilize on the inner index within the time cycle specified, after power has been applied.

3.14.3 *Circuit breaker.*

3.14.3.1 *Breaker open.* With 28 plus or minus 0.5 volts applied to the motor, with the two indexes separated by at least 200 yards, the output shaft for the ballistics drive locked (immovable), the circuit breaker shall open in 90 seconds or less. The check for this requirement shall be started when

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the motor is cold (standard ambient temperature). This check shall be followed with that of 3.14.3.2.

TABLE III. *Time of Response*

Position of outer index	Position of inner index	Voltage (Plus or minus 0.5V)	Total Elapsed Time
4,800 yds.	500 yds.	28 V.	4 seconds
500 yds.	4,800 yds.	28 V.	4 seconds
500 yds.	4,800 yds.	18 V.	8 seconds
4,800 yds.	500 yds.	18 V.	8 seconds

3.14.3.2 *Breaker closed.* At standard ambient temperature with the load specified in 3.12 applied to the output shaft for the ballistics drive, the computer operating at 28 plus or minus 0.5 volts, using the AP–T, M318A1 Cam, the input shaft shall be oscillated for a period of forty-five seconds over a spread of 1,000 yards plus or minus 100 yards between ranges of 1,200 to 2,800 yards at a rate of four to four and one-half revolutions per second, during which time the circuit breaker shall not open.

3.15 Hand crank and input shaft travel.

3.15.1 The rotation of the superelevation hand crank within the limits of its stops shall cover the excursion from minus 1.5 plus or minus 0.1 mils to plus 98.5 plus or minus 0.5 mils, as indicated on the superelevation counter when the ballistic correction knob is set at zero.

3.15.2 The range input shaft shall rotate through a total excursion of 24.0 plus 0.1 revolutions and shall cover the range from 480 to 5,000 yards, within this excursion, as measured from the 1,500 yard check point of the input shaft.

Information only

From 1,500 yards, 11R–218.4° .... ccw–480 yards
From 1,500 yards, 12R–94.6° .... cow–5,000 yards

3.16 Superelevation cams. The HVAP, M304 Cam shall be placed in the computer and checked, after which this cam shall be removed and placed in the spare cam box. Six cams shall then be placed in the computer in the following sequence starting with the AP–T, M318A1 and following with APC–T, M82; HEAT, T108; HEP, T142E3, HVAP, M332; and HE, M71. Each computer shall be shipped with the same seven cams with which it qualified.

3.17 Insulation breakdown and resistance.

3.17.1 Insulation breakdown tests shall be in accordance with Specification MIL–F–13926 except that breakdown voltage of 500 volts root mean square shall be substituted for 750 volts RMS.

3.17.2 Insulation resistance tests shall be in accordance with Specification MIL–F–13926 except that the resistance value shall not be less than 2 megohms and a direct current potential of 500 volts shall be used in place of 800 volts. All lamps shall be removed from sockets prior to tests.

3.18 Continuity. The electrical circuits shall be tested for continuity, and shall conform to the wiring diagram. When the hand crank is engaged the interrupter switch shall open the motor circuit and energize the pilot light. Pressing of the reset button shall cause the motor circuit to be closed and the pilot light to be extinguished. Actuation of the ammo selector handle shall cause the ammo selector interruption switch to open the motor circuit. Resetting of the handle shall cause switch to complete the circuit.

3.19 Illumination. The internal lighting device shall provide adequate illumination so that the inner and outer indexes of the range dial and the ammunition index shall be clearly defined and readable when observed in a darkened enclosure.

3.20 Identification marking. Identification marking, unless otherwise specified shall be in accordance with Standard MIL–STD–130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Gov-

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TABLE IV

Computer accuracy as determined by position of rotation of the output shaft based on 1,500 yard reference range for AP-T, M318A1 Cam (Dwg #D8585366) Informational only *Corresponding nominal superelevation values in mils.

Range	500 yds	1,000 yds	1,500 zero check	2,000 yds	3,000 yds	3,600 yds	4,000 yds	4,500 yds
Position of input shaft.	11R 68.8° CW	4R 46.9° CW	0° Check Point	2R 334.6° CCW	7R 21.6° CCW	8R 330° CCW	9R 356.5° CCW	11R 305° CCW
Input shaft indexing error.	±1°	±1°	±1°	±1°	±1°	±1°	±1°	±1°
AP-T, M318A1.	CCW 1R89.4° *2.78	CW 232.6° *5.79	0° *9.02	CW 249.5° *12.49	CW 2R87.2° *20.24		CW 4R23.1° *29.35	CW 5R284.2° *37.97
Heat T108.	CCW 1R50.5° *3.32	CCW 138.1° *7.11	CW 174.4° *11.45	CW 1R175.4° *16.47	CW 4R7.3° *29.13		CW 7R188.2° *46.64	CW 11R105.1° *65.49
HEP, T142E3.	CCW 1R3.5° *3.97	CCW 23.1° *8.70	CW 1R33.6° *14.50	CW 2R190.4° *21.67	CW 6R184.2° *41.59	CW 9R294.3° *58.11		
HVAP, M332.	CCW 1R182.3° *1.49	CCW 1R28.2° *3.63	CCW 228.5° *5.85	CCW 49.1° *8.34	CW 1R37.9° *14.54		CW 2R314.3° *23.39	CW 4R333.3° *33.66
HE, M71 (REN) 2,700 F/S.	CCW 1R37.5° *3.50	CCW 132.4° *7.18	CW 147.6° *11.08	CW 1R86.3° *15.23	CW 3R25.5° *24.38		CW 5R63.5° *34.90	CW 7R30.6° *44.45
APC-T, M82.	CCW 1R59.3° *3.20	CCW 175.4° *6.59	CW 84.6° *10.20	CW 1R2.3° *14.06	CW 2R256.5° *22.59		CW 4R243.2° *32.40	CW 6R165.1° *41.32
HVAP, M304.	CCW 1R124.4° *2.29	CCW 302.3° *4.82	CCW 101.4° *7.61	CW 119.6° *10.69	CW 1R280.4° *17.91	CW 2R294.9° *23.11		
Tolerance output shaft	10.75° *0.15	10.75° *0.15	7.25° *0.10	14.5° *0.20	21.5° *0.30	28.75° *0.40	36° *0.50	50.5° *0.70

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ernment. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. Inspection requirements and procedures shall be in accordance with Specification MIL–I–45208.

4.3 Initial production sample. Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 *Initial production sample failure.* Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation. Inspection lot sizes and lot formation shall be in accordance with Standard MIL–STD–105.

4.4.1 *Defective units and lots.* Defective units and lots shall be processed as specified in Standard MIL–STD–105.

4.5 Material. Control of material shall be in accordance with test data requirements of Specification MIL–I–45208.

4.6 Inspection of components. Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed on Supplementary Quality Assurance Provisions List (LSQAP) 8585-600.

4.6.1 *Contents of SQAP.* The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of each component covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual nonconforming items.

4.6.2 *Contractor's inspection for all components not covered by AQAP.* Inspection for these components shall be in accordance with the inspection procedures of the inspection system required by Specification MIL–I–45208. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the computer.

4.7 Inspection equipment. Inspection equipment shall conform to drawings listed in IEL–8585600, except as provided for under the alternative inspection procedures and inspection equipment requirements of Specification MIL–I–45208. Provisions of Specification MIL–I–45607 shall apply to test equipment provided by the Government or provided by the contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired, for verification inspection.

4.7.1 *Accuracy requirements.* Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test

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equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8 **Acceptance inspection.**

4.8.1 *Classification of defects.* The classification of defects in tables V and VI shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lot. The AQL specified shall apply to each individual characteristic as a basis for acceptance or rejection.

4.8.2 *Disposition of rejected lots.* Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

4.9 **Test methods and procedures.**

4.9.1 *Storage temperature.* The temperature cycling for the physical endurance test of the computer at minus 80° and plus 160° Fahrenheit shall be conducted in accordance with Procedure II of Specification MIL-F-13926. Upon return to standard ambient temperature, the computer shall be subjected to a visual and tactile examination and shall show no evidence of physical failure. The computer shall then be subjected to the test in 4.9.3 and shall conform to the requirements specified.

4.9.2 *Operational temperature.* The temperature cycling for the operational test of the computer at the specified temperature extremes shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond minus 40° Fahrenheit or raised beyond plus 150° Fahrenheit. While the computer is thermally stabilized at each temperature extreme it shall be subjected to the applicable tests in 4.9.2.1 through 4.9.2.3 inclusive.

4.9.2.1 *Operating torques and force, minus 40° and plus 150° Fahrenheit.* When thermally stabilized at each temperature extreme the controls of the computer shall be subjected to the tests specified in 4.9.8. The torque or force required to operate the controls at these temperature extremes shall not exceed the applicable value specified in Table II.

4.9.2.2 *Outer index and inner index, minus 40° Fahrenheit.* When the computer is thermally stabilized at minus 40° Fahrenheit apply the specified voltage and place the circuit breaker switch in the ON position. By means of a continuous rotation of the input shaft from the clockwise (cw) stop to the counterclockwise (ccw) stop traverse the inner index over the entire span of the range scale within the time limit specified. Continue test by rotating the input shaft from the ccw stop to the cw stop. During this excursion of the inner index the outer index shall meet the requirements of 3.5.2 (b).

4.9.2.3 *Outer index and inner index, plus 150° Fahrenheit.* When the computer is thermally stabilized at plus 150° Fahrenheit apply the specified voltage and place the circuit breaker switch in the OFF position. Set the ammunition selector on the cam specified, the range correction knob to zero, apply the specified torque load to the computer output shaft, position the computer inner index at 500 yards and the outer index at 4,500 yards on the range scale. When the circuit breaker switch is thrown to the ON position, the outer index shall "settle out" (cease oscillatory operation) coincident to the inner index within eight changes (click) in direction of the computer servo. The computer shall again meet this requirement when the test is repeated with the outer index positioned at 500 yards and the inner index positioned at 4,500 yards on the range scale.

4.9.3 *Post environmental check.* Upon return to standard ambient temperature the computer shall be set up as specified in 4.9.7 and subjected to the following tests:

(a) *Stability.* This test shall be conducted as specified in 4.9.7.7 to

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TABLE V. *Classification of Defects*
Use Inspection Level II in table III–A with Sampling Plan table IV–A
of Standard MIL–STD–105

Critical. None defined.

	AQL 1.0 percent defective	Requirement	QA. par.	Test method
Major:				
101.	Physical failure, post vibration	3.9	4.9.6	Visual-Tactile.
102.	Output shaft error (manual)	3.10.2	4.9.7.1	Special-Testing Equipment (STE).
103.	Outer index and inner index error	3.10.3	4.9.7.2	STE.
104.	Backlash, excessive	3.10.3.1	4.9.7.3	STE.
105.	Outer index and range scale error	3.10.4	4.9.7.4	STE.
106.	Servo followup inaccurate	3.11	4.9.7.5	STE-Audio.
107.	Output shaft error (power)	3.12	4.9.7.6	STE.
108.	Computer servo unstable	3.13	4.9.7.7	STE-Audio.
109.	Time of response inadequate	3.14.2	4.9.7.8	Standard Measuring Equipment (SME)-STE.
110.	Circuit breaker malfunction	3.14.3	4.9.7.9	STE-SME.
111.	Hand crank rotation inadequate	3.15.1	4.9.7.10	Tactile-Visual.
112.	Input shaft travel incorrect	3.15.2	4.9.7.11	STE.
113.	Cam arrangement incorrect	3.16	4.9.7.12	STE-Visual.
114.	Operating torques and force excessive	3.14.1	4.9.8	STE-SME.
115.	Insulation breakdown	3.17.1	4.9.9	SME.
116.	Insulation resistance inadequate	3.17.2	4.9.10	SME.
117.	Lack of continuity	3.18	4.9.11	SME-Tactile.
118.	Illumination inadequate	3.19	4.9.12	Visual.
119.	Identification markings incorrect	3.20	4.9.13	Visual.

Minor: None defined.

Note. The above tests shall be conducted at standard ambient temperature (plus 60° to plus 90° F.) in accordance with the temperatures listed.

TABLE VI. *Classification*
Use Inspection Level L7 in table III–B with Sampling Plan table IV–A
of Standard MIL–STD–105

Critical: None defined.

	AQL 4.0 percent defective	Requirement	QA. par.	Test method
Major:				
101.	Physical failure	3.5.1	4.9.1	Visual-Tactile.
102.	Operating torques and force excessive	3.5.2(a)	4.9.2.1	Special Test Equipment (STE)-Standard Measuring Equipment (SME).
103.	Outer index and inner index error	3.5.2(b)	4.9.2.2	STE.
104.	Outer index and inner index error	3.5.3(c)	4.9.2.3	STE.
105.	Computer servo unstable	3.6(a)	4.9.3(a)	STE-Audio.
106.	Operating torque and force excessive	3.6(a)	4.9.3(b)	STE-SME.
107.	Time of response inadequate	3.6(a)	4.9.3(c)	STE-SME.
108.	Circuit breaker malfunction	3.6(a)	4.9.3(d)	STE-SME.
109.	Output shaft error (power)	3.6(b)	4.9.3.(e)	STE.
110.	Leakage (sealing)	3.7	4.9.4	STE.
111.	Drying inadequate	3.8	4.9.5	Visual.

Minor: None defined.

Note. The tests for characteristic 102 shall be conducted at minus 40° and plus 150° F., respectively. The test for characteristic 103 shall be conducted at minus 40° F. The test for characteristic 104 shall be conducted at plus 150° F. The test for characteristic 101 and 105 through 111 inclusive shall be conducted at standard ambient temperature subsequent to the storage and operability test at the temperature extremes.

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determine compliance with the requirements of 3.13.

(b) *Torque and force.* This test shall be conducted as specified in 4.9.8. The torque and force values obtained shall not exceed the values specified in Table II for standard ambient temperature.

(c) *Time of response.* This test shall be conducted as specified in 4.9.7.8. The computer shall meet the requirements of 3.14.2.

(d) *Circuit breaker.* This test shall be conducted in accordance with 4.9.7.9. The circut breaker shall meet the requirements of 3.14.3.

(e) *Power operation.* This test shall be conducted as specified in 4.9.7.6. The rotational position of the output shaft shall meet the requirements of 3.6(b).

4.9.4 *Sealing.* This test shall be conducted in accordance with Procedure I of Specification MIL–F–13926. The computer shall show no evidence of water penetration as a result of this test. The Receptacle 8200970 shall be covered with water-proof tape during this test. Subsequent to the interior inspection for water penetration the computer shall be resealed.

4.9.5 *Drying.* The computers subjected to the test specified in 4.9.1, 4.9.2 and 4.9.4 shall be internally air dried as required in 3.8 (see 6.5).

4.9.6 *Vibration.* This operation shall be performed on Fixture 7650085 using Adapter 7659608. Position the ammunition selector and outer index as specified. The computer shall be vibrated for five minutes plus or minus fifteen seconds. Subsequent to vibration the computer shall be subjected to a tactile and visual examination and it shall show no evidence of damage or evidence of broken, loose or missing parts. The computer shall then be subjected to the tests in 4.9.7 through 4.9.12 inclusive and shall meet the requirements specified.

4.9.7 *Orientation for test.* The following tests shall be conducted with the computer positioned on Test Fixture 8565298 as specified on Drawing F8565296. The computer input shaft and the fixture input mechanism shall be oriented to the 150 yard check point as outlined in the setup instructions on Drawing F8565296.

4.9.7.1 *Output shaft.* Set the ammunition selector on the cam specified and the range correction knob on zero. This test shall be conducted in accordance with paragraph 4.2.1 on Drawing F8565296. When the mil value specified in table I of 3.10.2 is indicated on the computer superelevation mil counter, the rotational position of the output shaft for the ballistics drive shall not exceed the tolerance specified in table I.

4.9.7.2 *Outer index and inner index.* Set the ammunition selector on the cam specified and the range correction knob on zero. This test shall be conducted in accordance with paragraph 4.2.2 on Drawing F8565296. When the rotational position of the computer input shaft corresponds to the value for each of the ranges indicated in table IV and the computer outer and inner index are coincident, the rotational position of the ballistics drive output shaft shall not exceed the tolerance specified for the applicable value in table IV. The remaining six cams shall meet the requirements of table IV for the specified range.

4.9.7.3 *Backlash.* This test shall be conducted in accordance with paragraph 4.2.3 on Drawing F8565296. Backlash of the computer superelevator output shaft shall not exceed the tolerance specified in 3.10.3.1 when measured at four points of shaft rotation. The four points of measure shall be approximately 90° apart.

4.9.7.4 *Outer index and range scale.* Set the ammunition selector on the cam specified and the range correction knob on zero. This test shall be conducted in accordance with paragraph 4.2.4 on Drawing F8565296. When the computer outer index is coincident to the selected range mark on the computer range scale, the rotational position of the output shaft for the ballistics drive shall not exceed the tolerance specified for the applicable value in table IV.

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4.9.7.5 *Servo followup sensitivity.* Set the ammunition selector on the cam specified, the range correction knob on zero, and the inner and outer index to a range between 4,000 yards and 4,800 yards. This test shall be conducted in accordance with paragraph 4.2.5 on Drawing F8565296. With the computer in power operation, the average and maximum recorded change (closing of computer servo contacts) resulting from a total rotation of the output shaft 72° clockwise and 72° counterclockwise shall not exceed the tolerance specified in 3.11. Superelevation increment changes shall be measured at the computer output shaft for the ballistics drive.

4.9.7.6 *Power operation.* This test shall be conducted as specified in paragraph 4.2.6 on Drawing F8565296 with the range correction knob set to zero.

4.9.7.6.1 With the input voltage set at 28 plus or minus 0.5 volts and the circuit breaker switch in the ON position, the accuracy of each cam shall be tested at the ranges listed in table IV for computer output shaft rotation. When the input shaft is rotated to a value indicated in table IV for a given range, the rotational position of the output shaft shall correspond to the value for that range in table IV within the tolerance specified.

4.9.7.6.2 With the input voltage set at 18 plus or minus 0.5 volts, the computer shall meet the accuracy requirements of table IV for the cams and ranges specified in 3.12.

4.9.7.7 *Stability.* Prepare the computer for this test in accordance with paragraph 4.2.7 on Drawing F8565296, set the range correction knob to minus fifteen percent within one percent and adjust the input voltage to 28 plus or minus 0.5 volts. Rotate the computer inner index 1,500 yards, throw the circuit breaker switch to ON position and energize the servo motor. The outer index shall "settle out" (cease oscillatory operation) coincident to the inner index within eight changes (clicks) in direction of the computer servo. The inner index shall then be rotated to four different ranges between 1,500 yards and minimum range, and the outer index shall again "settle out" coincident to the inner index within eight changes in direction of the servo. Set the range correction knob to plus fifteen percent within one percent and repeat the aforementioned test. The six cams installed in the computer shall be subjected to this test.

4.9.7.8 *Time of response.* Set the ammunition selector on the cam specified and the range correction knob on zero. This test shall be conducted in accordance with paragraph 4.2.8 on Drawing F8565296. When the circuit breaker switch is thrown on the ON position and the servo motor is energized, the computer outer index shall stabilize on the inner index within the time cycle specified in table III of 3.14.2.

4.9.7.9 *Circuit breaker.*

4.9.7.9.1 *Opening.* Set the ammunition selector on the cam specified, the range correction knob on zero. Separate the inner and outer index the required distance. This test shall be conducted in accordance with paragraph 4.2.9.2 on Drawing F8565296. When the circuit breaker switch is thrown to the ON position and the servo motor is energized, the circuit breaker shall open (switch move to OFF position) within the time period specified in 3.14.3.1.

4.9.7.9.2 *Nonopening.* Set the ammunition selector and range correction knob as specified in 4.9.7.9, and the inner and outer index coincident on 2000 yards of range. This test shall be conducted in accordance with paragraph 4.2.9.1 on Drawing F8565296. The computer input shaft shall be oscillated as specified in 3.14.3.2 with the circuit breaker switch in the ON position and the servo motor energized. The circuit breaker shall not open (switch move to OFF position) as a result of this oscillation.

4.9.7.10 *Hand crank.* The test for hand crank travel shall be conducted in accordance with paragraph 4.2.10.1 on Drawing F8565296 and shall meet the requirements of 3.15.1.

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4.9.7.11 *Input shaft travel.* The test for input shaft travel shall be conducted in accordance with paragraph 4.2.10.2 on Drawing F8565296 and shall meet the requirements of 3.15.2.

4.9.7.12 *Superelevation cams, arrangement.* Subsequent to the inspection of the required cams they shall be positioned in the computer as specified. The process of placing or replacing the cams in the computer shall be as specified in paragraph 4.2.11 on Drawing F8565296.

4.9.8 *Operating torques and force.* The controls listed in table II of 3.14 shall not exceed the applicable torque and force values specified for the temperature indicated. The equipment to be utilized for this test is listed on the IEL. The reset button shall be inspected by use of standard measuring equipment. This standard measuring equipment shall be capable of measuring the values within one-quarter pound.

4.9.9 *Insulation breakdown.* This test shall be performed at any time subsequent to vibration. Remove lamps and disconnect capacitor from computer circuit prior to test. This test shall be conducted in accordance with Specification MIL–F–13926. During application of the test voltage there shall be no disruptive discharge or deterioration due to current leakage.

4.9.10 *Insulation resistance.* The conditioning of the computer for test shall be as specified in 4.9.9. This test shall immediately follow the insulation breakdown test and shall be conducted in accordance with Specification MIL–F–13926. Insulation resistance shall meet the requirements of 3.17.2.

4.9.11 *Continuity.* This test shall be performed subsequent to the tests specified in 4.9.9 and 4.9.11. The electrical circuits shall conform to the wiring diagram. Actuation of the hand crank, reset button and ammunition selector handle shall cause the computer to function as required in 3.18.

4.9.12 *Illumination.* The illuminated inner and outer indexes of the range dial and ammunition index of the computer shall meet the requirements of 3.19 when observed in a darkened enclosure with the unaided eye.

4.9.13 *Identification markings.* Markings shall be visually inspected for conformance to Standard MIL–STD–130.

4.9.14 *Packaging, packing and marking.* Quality assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL–C–45128.

4.10 **Final acceptance.** Final acceptance of the lot of computers shall be withheld by the Government inspector until the sample lot of computers, representative of that production lot, have satisfactorily passed the applicable requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 **Packaging, packing and marking.** Packaging, packing and marking shall be in accordance with Specification MIL–C–45128.

6. NOTES

6.1 **Intended use.** The Computer, Ballistics: M13A1 is used to receive the log of range data from a Range Finder M13A1 calibrated for range measurement in yards, and converts the value to superelevation for various ammunitions, and transmits same to the gun sighting system in the 90-mm Gun Tanks M48, M48A1 and M48A2.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of an applicable level of preservation, packaging and packing

(c) Applicable stock number

(d) Provisions for initial production sample including data of delivery (see 3.1)

6.3 **Design information.** Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

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6.4 Definitions.

6.4.1 *Standard measuring equipment (SME).* Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.4.2 *Special test equipment (STE).* Special test equipment is defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.5 Drying interior of computer. Computers subjected to the requirements of 3.5 and 3.7 should be dried when the top of the computer is removed for inspection to determine acceptance to the requirements of 3.7. Should the computer be subjected only to the requirements of 3.5 the cam change plate may be removed to facilitate compliance with the requirements of 3.8.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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